the group as a whole; or selfish, practised for the good of the individual doing the reproducing. I will deal with the two theories in order.

Wynne-Edwards supposed that individuals have fewer children than they are capable of, for the benefit of the group as a whole. He recognized that normal natural selection cannot possibly give rise to the evolution of such altruism: the natural selection of lower-than-average reproductive rates is, on the face of it, a contradiction in terms. He therefore invoked group selection, as we saw in Chapter 1. According to him, groups whose individual members restrain their own birth rates are less likely to go extinct than rival groups whose individual members reproduce so fast that they endanger the food supply. Therefore the world becomes populated by groups of restrained breeders. The individual restraint that Wynne-Edwards is suggesting amounts in a general sense to birth control, but he is more specific than this, and indeed comes up with a grand conception in which the whole of social life is seen as a mechanism of population regulation. For instance, two major features of social life in many species of animals are *territoriality* and *dominance hierarchies*, already mentioned in Chapter 5.

Many animals devote a great deal of time and energy to apparently defending an area of ground which naturalists call a territory. The phenomenon is very widespread in the animal kingdom, not only in birds, mammals, and fish, but in insects and even sea anemones. The territory may be a large area of woodland which is the principal foraging ground of a breeding pair, as in the case of robins. Or, in herring gulls for instance, it may be a small area containing no food, but with a nest at its centre. Wynne-Edwards believes that animals who fight over territory are fighting over a *token* prize, rather than an actual prize like a bit of food. In many cases females refuse to mate with males who do not possess a territory. Indeed it often happens that a female whose mate is defeated and his territory conquered promptly attaches herself to the victor. Even in apparently faithful monogamous species, the female may be wedded to a male's territory rather than to him personally.

If the population gets too big, some individuals will not get territories, and therefore will not breed. Winning a territory is therefore, to Wynne-Edwards, like winning a ticket or licence to breed. Since there is a finite number of territories available, it is as if a finite number of breeding licences is issued. Individuals may fight over who gets these licences, but the total number of babies that the population can have as a whole is